

1 Fig. 1.1 shows a hinged beam of length 60.0 cm held horizontally against a wall by a cord XY.

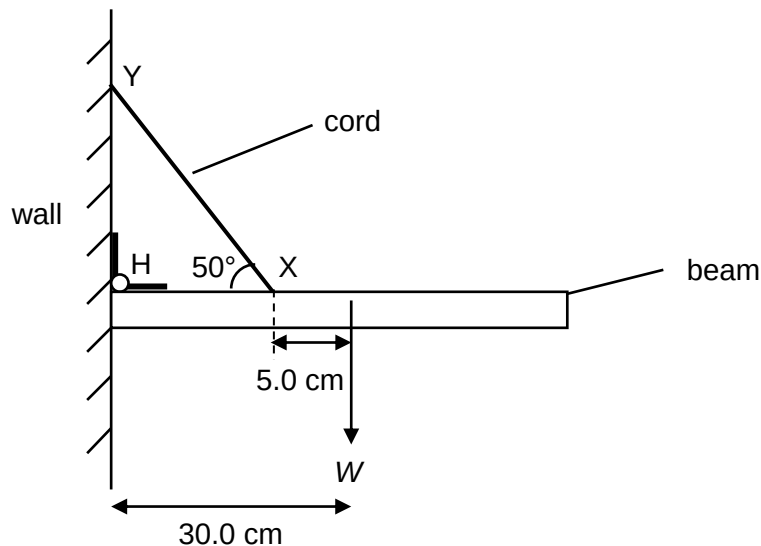


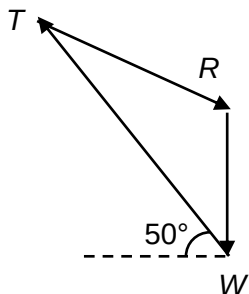
Fig. 1.1

The forces acting on the beam are its weight  $W$ , the force  $T$  exerted by the cord, and the force  $R$  exerted by the hinge  $H$ .

(a) In the space provided below, sketch a labelled vector triangle of the forces acting on the beam.

[2]

Solution:



- Correct labelling of forces and arrows point in the same direction (closed polygon).
- $R$  must be diagonally down not up, i.e. must have both a rightward & downward component.

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|  | <b>(b)</b> The weight of the uniform beam is 40.0 N and the mass of the cord is negligible<br>Calculate the magnitude of                                                                                                                                                                                                                    |
|  | <b>(i)</b> the tension $T$ ,                                                                                                                                                                                                                                                                                                                |
|  | $T = \dots\dots\dots$ N [2]                                                                                                                                                                                                                                                                                                                 |
|  | By principle of moments,<br>taking moments about the hinge H,<br>Sum of clockwise moment = sum of anti-clockwise moment<br>$40.0 \times 0.300 = T \sin 50^\circ \times 0.250$<br>$T = 62.7$ N<br><br>1<br>1                                                                                                                                 |
|  |                                                                                                                                                                                                                                                                                                                                             |
|  | <b>(ii)</b> the force $R$ .                                                                                                                                                                                                                                                                                                                 |
|  | $R = \dots\dots\dots$ N [2]                                                                                                                                                                                                                                                                                                                 |
|  | Applying the cosine rule for the vector triangle,<br>$R^2 = T^2 + W^2 - 2TW \cos 40^\circ$<br>$R^2 = 62.7^2 + 40.0^2 - (2 \times 62.7 \times 40.0 \times \cos 40^\circ)$<br>$R = 41.1$ N<br><br>1<br>1<br><br>Method 2:<br>Setting up 2 equations for Vertical & Horizontal Equilibrium, and then solving these 2 equations simultaneously. |
|  | <b>(c)</b> A brick is placed on the beam at X without the cord snapping. Subsequently, when the brick is shifted further away from the hinge along the beam, the cord snaps.<br><br>Explain why the cord snaps.                                                                                                                             |
|  | .....                                                                                                                                                                                                                                                                                                                                       |
|  | .....                                                                                                                                                                                                                                                                                                                                       |
|  | .....                                                                                                                                                                                                                                                                                                                                       |
|  | ..... [2]                                                                                                                                                                                                                                                                                                                                   |
|  | The <u>total clockwise moments about the hinge H is larger than the total anti-clockwise moment</u> that can be provided by the tension in the cord.<br><br>Hence the <u>cord snaps as the tension <math>T</math> has exceeded its breaking strength.</u><br><br>1<br>1                                                                     |
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| 2 | <b>(a)</b> State the principle of conservation of linear momentum |
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